

at the time, not as seriously as I would do it now. It was decided that the way women and doctors used the key covariates was pretty much the same around the country, and any differences were probably captured by the observed covariates.

Propensity scores were estimated by logistic regression, and they were used to create five subclasses of treatment/control women. The women were ranked by their estimated propensity scores, and the lowest 20% formed subclass 1, the next 20% formed subclass 2, etc. Within each subclass, balance was checked, not only on the covariates included in the propensity score, but also on all other important covariates in the database. For example, the average age of a treated women within each subclass should be approximately the same as the average age of a control women in that subclass, and the proportion of each that are married should also be as similar as if the treatment and control women in that subclass had been randomly divided (obviously, not with equal probability across the subclasses). When less balance was found on a key covariate within a subclass than would have occurred in a randomized experiment, terms were added to the propensity score model and balance was reassessed. Unfortunately, those tables and the processes never survived into the final report, but such balance was achieved—not perfectly, but close enough to believe in the hypothetical underlying randomized block experiment that led to the observed data.

The results of the subclassification on the propensity score are summarized in Table 2. In general, this observational study’s results are consistent with those from the randomized trials. There is essentially no evidence for any advantage to the radical operation, except possibly in those propensity score subclasses where the women and doctors were more likely to select

TABLE 2

Estimated 5-year survival rates for node-negative patients in SEER data base within each of five propensity score subclasses: from tables in U.S. GAO Report [General Accounting Office (1994)]

Propensity score subclass	Treatment condition	<i>n</i>	Estimate
1	Brest conservation	56	85.6%
	Mastectomy	1008	86.7%
2	Brest conservation	106	82.8%
	Mastectomy	964	83.4%
3	Brest conservation	193	85.2%
	Mastectomy	866	88.8%
4	Brest conservation	289	88.7%
	Mastectomy	978	87.3%
5	Brest conservation	462	89.0%
	Mastectomy	604	88.5%

mastectomy (subclasses 1, 2, 3), but the data are certainly not definitive. Similarly, for the women and doctors relatively more likely to select breast conserving operations, there is some slight evidence of a survival benefit to that choice. If we believed that the treatment effect should be the same for all women in the study, these changing results across propensity subclasses could be viewed as evidence of a confounded and nonignorable treatment assignment (i.e., an omitted key covariate). Overall, however, there appears to be no advantage to recommending one treatment over the other. It is interesting to note that, consistent with expectations, the overall survival rates in the observational dataset are not as good as those in the more specialized centers represented in Table 1.

4.4. Marketing example. Propensity score methods, like randomization, work best in large samples. For a trivial example, if we have one man and one woman, one to be treated and one to be control, randomized treatment assignment in expectation would create a half-man treated and half-woman control, but in reality the man would be either treated or control and the woman would be in the other condition. With a hundred men and a hundred women, we would expect roughly half of each to be in each treatment arm. Analogously, with propensity scores, the creation of narrow subclasses or matched pairs should create balanced distributions in expectation, which should be easier to achieve and to assess in large samples than in small ones.

The next example, from [Rubin and Waterman \(2006\)](#), illustrates this feature well because the sample sizes are large: 100,000 treated doctors and 150,000 control doctors. “Treated” here means visited by a sales representative (rep) at least once during a certain six month period; the sales rep tells the doctor the details of a new weight-loss drug being promoted by a pharmaceutical company. The control doctors are not visited by a sales rep from that company during that period. The treatment/control indicator variables come from the companies’ records as provided by the sales reps. The key outcome variable is the number of prescriptions (scripts) of this drug written by the doctor during the following six months; this information on scripts is obtained several months later from a third party vendor, which is updated at regular intervals. The previous version of this data source and other sources have all sorts of background information on the doctors, such as sex, race, age, years since degree, size of practice, medical specialty, number of scripts written in prior years for the same class of drugs as being described by the sales rep, etc. In fact, there are well over 100 basic covariates available. The objective of the observational study is to estimate the causal effects of the reps visiting these doctors. It costs money to pay the sales reps’ salaries to visit the doctors, and moreover, many reps get commissions based, in part, on the number of scripts written by the doctors they visited for the detailed

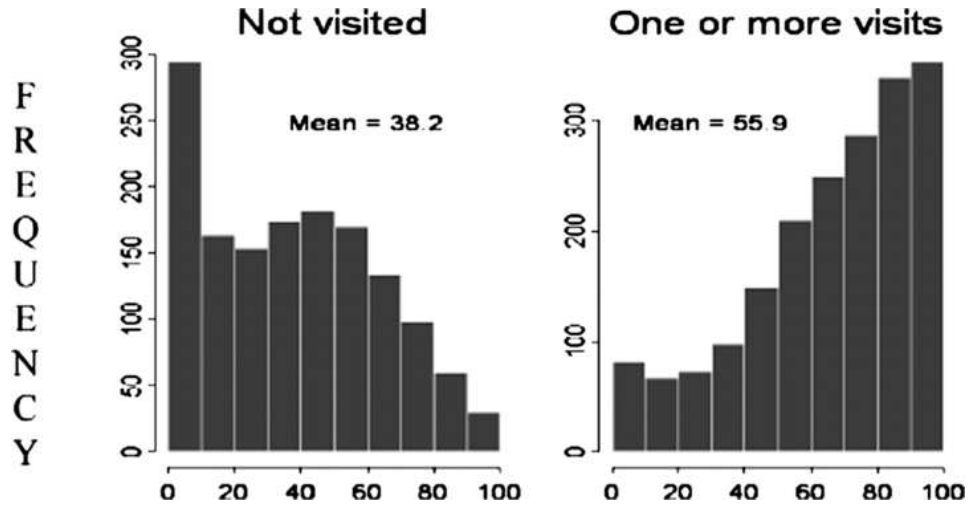


FIG. 1. Histograms for background variable: prior Rx score (0–100) at baseline.

drug. Do the visits cause more scripts to be written, and if so, which doctors should be visited with higher priority? Both of these, and other similar questions, are causal ones.

The decision-maker for visiting or not the doctors is essentially the sales rep, and these folks, rather obviously, like to visit doctors who prescribe a lot, who have large practices, are in a specialty that prescribes a lot of the type of drug being detailed, etc. Essentially all of these background variables, X , and more, are available on the purchased data set, which has huge sample sizes; the company has the indicator W for visited versus not, and next

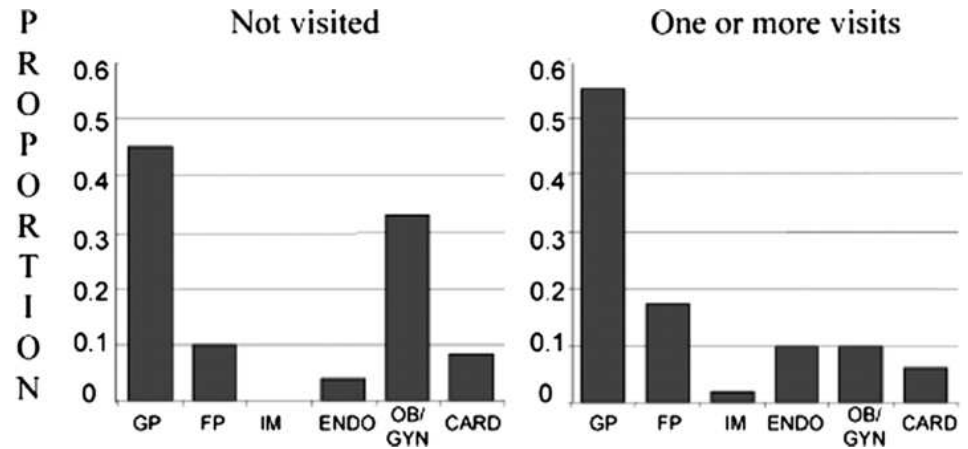


FIG. 2. Histograms for background variable: specialty.

year's purchased data set will have the outcome variables Y on the actual number of scripts written by these doctors in the next time period. So things look in good shape to estimate and re-estimate the propensity scores until we achieve balanced distributions within subclasses, or we decide that there are some types of doctors who have essentially no chance of being visited or not being visited, and then no estimation of causal effects will be attempted for them.

Figures 1 and 2 display the initial balance for two important covariates, number of prior scripts written in the previous year (for drugs in the same class as the detailed drug) on a scale from 0 (minimum) to 100 (the arbitrarily scaled maximum), and the specialty. These figures reveal quite dramatic differences between the doctors who were visited and those who were not visited. It is not surprising that the visited doctors were the ones who wrote many more prescriptions (per doctor) than the not visited doctors. But the visited doctors also have a different distribution of specialties than the not visited doctors. For example, ob-gyn doctors are visited relatively less often than doctors with other specialties; presumably, ob-gyn doctors do not prescribe weight-loss drugs for their pregnant patients, and the sales reps use this information.

Propensity scores were estimated by logistic regression based on various functions of all of the covariates. Figure 3 displays the histograms for the estimated linear propensity scores (the $\hat{\beta}X$ in the logistic regression) among the not visited and visited doctors. These histograms are shown with 15 subclasses (or bins) of propensity scores. In some bins, there are only visited doctors, that is, in the bins with linear propensity scores larger than 1.0; in those two bins, there are no doctors who were not visited. Presumably, they are high prescribing doctors with large practices, etc. No causal inferences are possible for them without making model-based assumptions relating outcomes to covariates for which there are no data to assess the underlying assumptions. Similarly, for the four lowest bins of propensity, with linear scores less than 0.1, all doctors are not visited, and so, similarly, no causal inferences about the effect of visiting this type of doctor are possible unless based on unassessable assumptions.

But in the other nine bins, there are both visited and not visited doctors, and the claim is that within each of those bins, the distributions of all covariates that entered the propensity score estimation will be nearly the same for the visited and not visited doctors. To be specific, let us examine the bin between 0.5 and <0.6 . Figures 4 and 5 show the distributions of prior number of prescriptions and specialties in this bin for the not visited and visited doctors. These distributions are strikingly more similar than their counterparts shown in Figures 1 and 2. In fact, they are so similar that one could believe that, within that bin, the visited doctors are a random sample from all doctors in that bin. And the claim is that this will hold (in